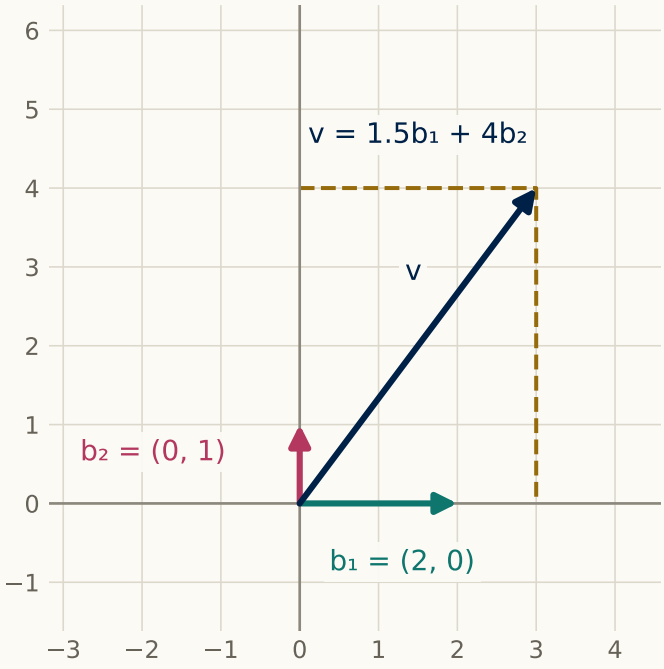Orthogonal rulers read independent coordinates